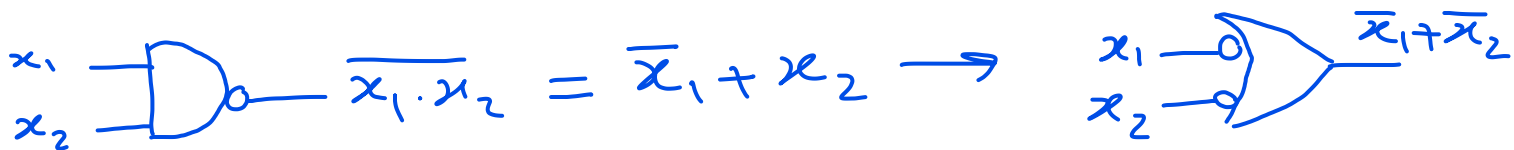


NAND & NOR Gates

- NAND and NOR gates are universal gates, meaning any digital circuit can be built using only one of them.
- In CMOS logic, NAND & NOR gates have simpler and faster transistor-level implementations than AND & OR gates because they require **fewer transistors** and avoid additional inverter stages.
- The **cost** of creating NAND & NOR is less than that for AND & OR gates.
- Any AND-OR network (SOP) can be implemented as a **NAND-NAND** network.
- Any OR-AND network (POS) can be implemented as a **NOR-NOR** network.

- NAND Gate



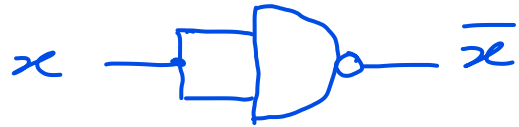
1. AND logic using NAND:



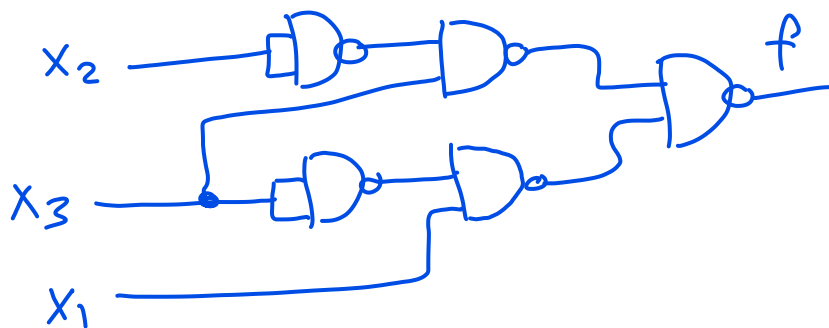
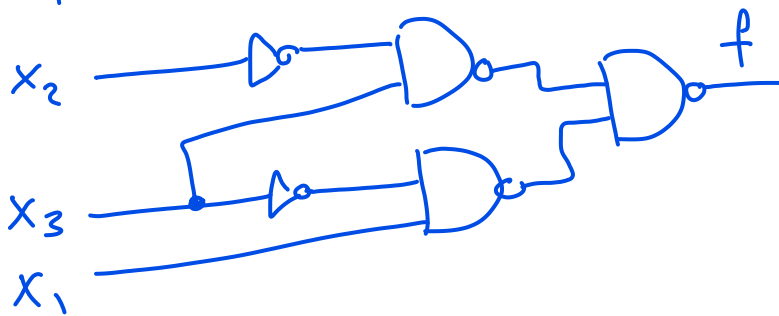
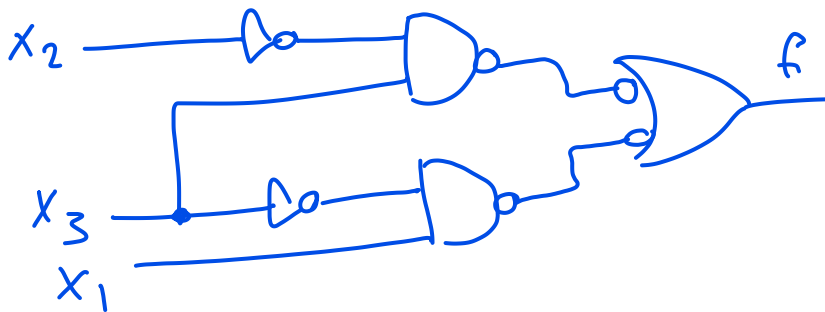
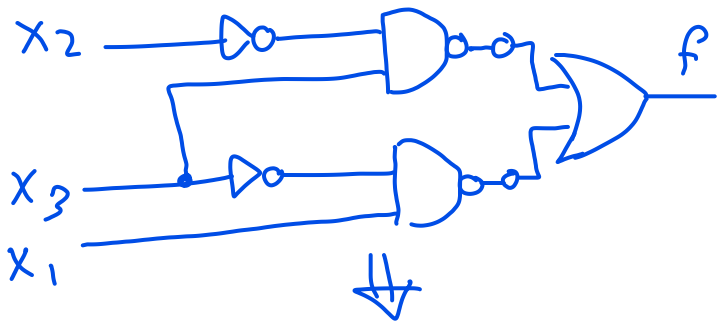
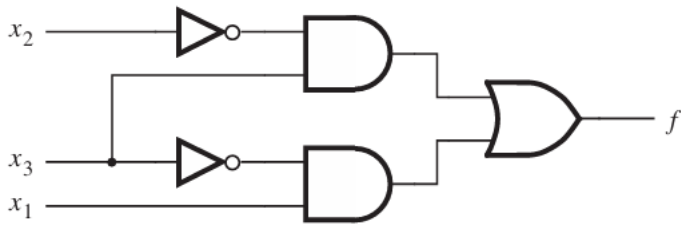
2. OR logic using NAND:



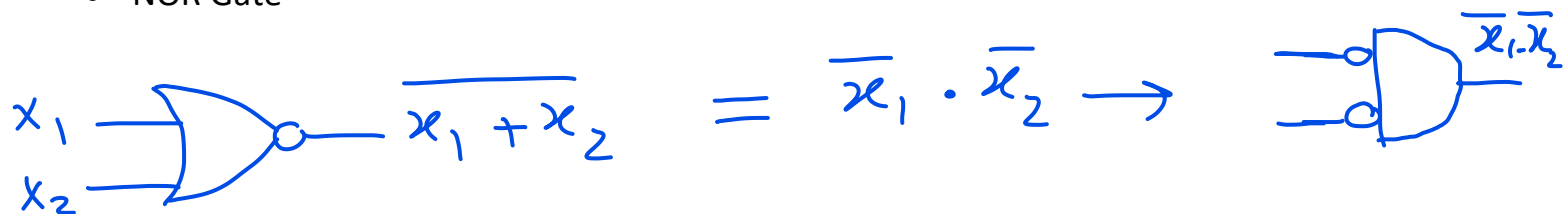
3. NOT logic using NAND



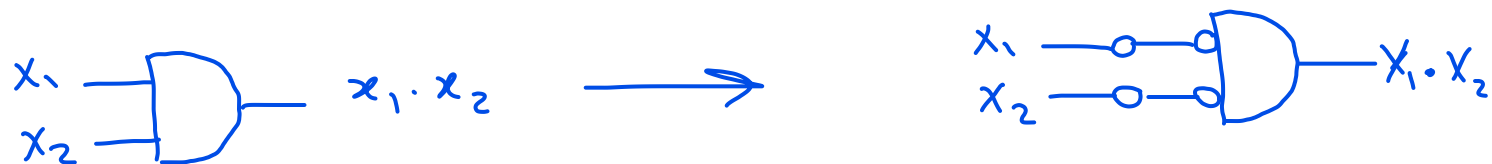
Example 1: Implement the following circuit using **only NAND** gates (i.e., NAND-NAND network).



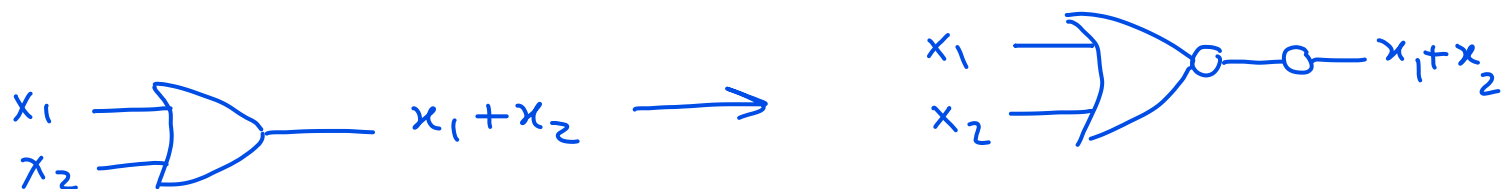
- NOR Gate



1. AND logic using NOR:



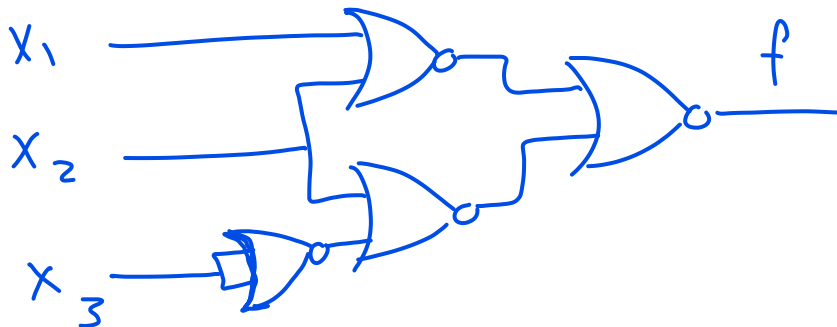
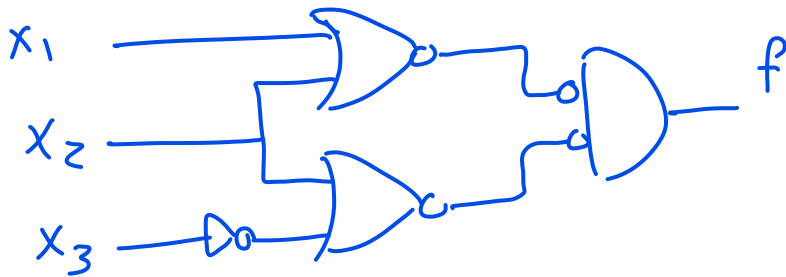
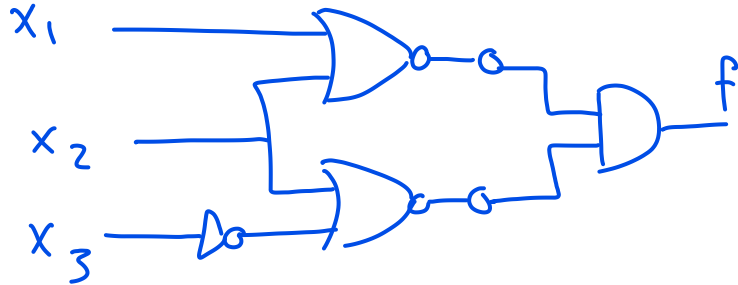
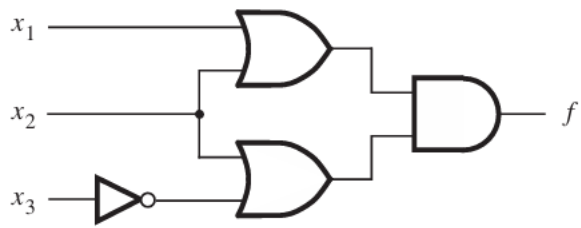
2. OR logic using NOR:



3. NOT logic using NOR



Example 2: Implement the following circuit using **only NOR** gates (i.e., NOR-NOR network).



Example: Draw the following circuit using only NOR.

